

Test ID	Category	Test being carried out	Test input / condition	Expected result
A-V01	Normal	Normal valid assessment	Score 15, max 20, pass mark 10	Valid record
A-V02	Score	Score equal to maximum	Score 20, max 20	Valid record
A-V03	Score	Score below pass mark	Score 5, max 20, pass mark 10	Valid record; represents a failed assessment
A-V04	Score	Score equal to zero	Score 0, max 20	Valid record
A-V05	Score	Score greater than maximum	Score 25, max 20	Invalid record
A-V06	Score	Negative score	Score -5	Invalid record
A-V07	Score	Non-numeric score	Score = abc	Invalid record
A-V08	Score	Missing score	Score left blank	Invalid record
A-V09	Max score	Negative maximum score	Max score -20	Invalid record
A-V10	Max score	Maximum score equal to zero	Max score 0	Invalid record
A-V11	Max score	Non-numeric maximum score	Max score = xyz	Invalid record
A-V12	Max score	Missing maximum score	Max score left blank	Invalid record
A-V13	Pass mark	Negative pass mark	Pass mark -5	Invalid record
A-V14	Pass mark	Pass mark greater than maximum	Pass mark 25, max 20	Invalid record
A-V15	Pass mark	Pass mark equal to maximum	Pass mark 20, max 20	Valid record
A-V16	Pass mark	Pass mark equal to zero	Pass mark 0	Valid record
A-V17	Pass mark	Non-numeric pass mark	Pass mark = pqr	Invalid record
A-V18	Pass mark	Missing pass mark	Pass mark left blank	Invalid record
A-V19	Date	Normal valid date	03/02/2026	Valid record
A-V20	Date	Old but valid assessment date	15/01/2020	Valid record
A-V21	Date	Invalid date format	not-a-date	Invalid record
A-V22	Date	Missing assessment date	Date left blank	Invalid record
A-V23	Date	Assessment date in future – first test	15/01/2027	Invalid record
A-V23-R	Date	Assessment date in future – retest after fix	15/01/2027	Invalid record
A-V24	Missing data	Missing subject	Subject left blank	Invalid record
A-V25	Missing data	Missing student ID	Student ID left blank	Invalid record
A-V26	Missing data	Missing assessment title	Title left blank	Invalid record
A-V27	Multiple invalid	Most assessment values missing	Main fields left blank	Invalid record
A-V28	Data type	Numeric fields contain text	Score jkl, max mno, pass mark pqr	Invalid record
A-V29	Data type	Mixed valid and invalid types	Score xyz, other values valid	Invalid record
A-V30	Null data	Fields contain NULL	Multiple fields contain NULL	Invalid record
AT-V01	Normal	Normal valid attendance	Total 100, attended 96	Valid record

AT-V02	Boundary	All sessions attended	Total 100, attended 100	Valid record
AT-V03	Boundary	Zero sessions attended	Total 100, attended 0	Valid record
AT-V04	Sessions attended	Attended greater than total sessions	Total 100, attended 120	Invalid record
AT-V05	Sessions attended	Negative sessions attended	Attended -5	Invalid record
AT-V06	Total sessions	Total sessions equal to zero	Total 0, attended 0	Invalid record
AT-V07	Total sessions	Negative total sessions	Total -100, attended 50	Invalid record
AT-V08	Data type	Non-numeric total sessions	Total = abc	Invalid record
AT-V09	Data type	Non-numeric sessions attended	Attended = xyz	Invalid record
AT-V10	Missing data	Missing total sessions	Total sessions left blank	Invalid record
AT-V11	Missing data	Missing sessions attended	Sessions attended left blank	Invalid record
AT-V12	Missing data	Missing student ID	Student ID left blank	Invalid record
AT-V13	Missing data	Both session values missing	Total and attended left blank	Invalid record
AT-V14	Null data	NULL total sessions	Total sessions = NULL	Invalid record
AT-V15	Null data	NULL sessions attended	Sessions attended = NULL	Invalid record
Processing tests				
A-P01	Assessment processing	Calculate improving grades	S001: 60%, then 70%	Average 13.0, change +2.0, previous failure 0
A-P02	Assessment processing	Detect previous failure	S002: 45%, then 40%, pass mark 50%	Average 8.5, change -1.0, previous failure 1
A-P03	Missing history	Student has only one assessment	S007 has fewer than 2 valid assessments	No ML features calculated; Insufficient data
AT-P01	Attendance processing	Calculate absences	Total 100, attended 96	4 absences
AT-P02	Attendance processing	Calculate attendance percentage	Total 100, attended 96	96.00%
AT-P03	Attendance processing	Calculate different absence counts	Normal records for S001–S010	Correct absence count for each student
Assessment_adapter ref				
R-01	Refactoring	Split assessment and attendance adapters into separate files	Move functions into assessment_adapter.py and process_assessments()	Existing program behaviour should remain unchanged
R-02	Refactoring	Remove duplicated assessment validation	changed to call	Assessment outputs and validation results should remain
R-03	Assessment validation	Full validation suite after refactor	Run all assessment validation tests again	All expected results should match actual results
R-04	Attendance validation	Full attendance validation suite	Run all 15 attendance tests	All expected results should match actual results
Merge testing				
M-01	Merge	Valid assessment and attendance	T001 has complete valid data	Ready
M-02	Eligibility	Insufficient assessment data	T002 has valid attendance but missing assessment features	Insufficient data
M-03	Eligibility	Missing attendance data	T003 has valid assessment but no attendance record	Insufficient data
M-04	Student retention	Student has neither assessment nor attendance	T004 exists only in student roster	Student retained and marked Insufficient data
M-05	Multiple subjects	Same student has two subjects	T005 has Maths and English with overall attendance of 3 absences	Two separate subject rows using same attendance
prediction testing				

M-01	Merge	Valid assessment and attendance data	T001 has valid Maths assessment features and valid	All four ML features merged and status set to Ready
M-02	Eligibility	Insufficient assessment data with valid attendance	T002 has valid attendance but missing assessment features	Student retained and status set to Insufficient data
M-03	Eligibility	Valid assessment data but missing attendance	T003 has valid Maths assessment features but no	Student retained and status set to Insufficient data
M-04	Student retention	Student has no assessment or attendance data	T004 exists only in student roster	Student should remain visible and be marked Insufficient data
M-05	Multiple subjects	One student has more than one subject	T005 has Maths and English assessments with 3 overall	Separate subject rows created and same attendance attached
P-01	Prediction	Ready student predicted as Not At Risk	S001 has complete model features	Model generates prediction and keeps status Ready
P-02	Prediction	Higher-risk sample passed into model	S002 has previous failure, grade average 8.5 and grade change -1	Model generates prediction using saved Random Forest
P-03	Prediction	Ready student predicted	S003 has complete valid model features	Model generates prediction
P-04	Prediction	Ready student predicted	S004 has complete valid model features	Model generates prediction
P-05	Prediction	Ready student predicted	S005 has complete valid model features	Model generates prediction
P-06	Prediction	Ready student predicted	S006 has complete valid model features	Model generates prediction
P-07	Prediction safety	Student with insufficient assessment history	S007 has attendance but missing required assessment features	Student must not be sent to model and should remain
P-08	Prediction safety	Student with no assessment data	S008 has attendance but no subject assessment features	Student must not be sent to model and should remain visible
P-09	Prediction safety	Student with no assessment data	S009 has attendance but no subject assessment features	Student must not be sent to model and should remain visible
P-10	Prediction safety	Student with no assessment data	S010 has attendance but no subject assessment features	Student must not be sent to model and should remain visible
P-11	Prediction verification	Compare expected prediction statuses with actual statuses	Expected statuses stored in prediction_test.csv for	Every expected status should match actual status
P-12	Prediction verification	Compare expected Random Forest outputs with actual	Expected predictions stored in prediction_test.csv for	Every expected prediction should match actual prediction

Data management t

S-01	Subject Management	Add valid subject	Enter Testing Maths and click Add Subject	Subject is added with a new SUB0010 ID
S-02	Subject Management	Prevent blank subject	Leave subject name blank	Error message shown and subject is not added
S-03	Subject Management	Prevent whitespace-only subject	Enter spaces only	Whitespace is removed, error shown. subject not added
S-04	Subject Management	Prevent duplicate subject	Enter existing subject Testing Maths	Duplicate error shown and subject is not added
S-04A	Subject Management	Prevent duplicate caused by excessive internal	Enter Te sting Maths when Testing Maths already exists	Extra whitespace should be normalised so accidental
S-04A	Subject Management	Prevent duplicate caused by excessive internal	Enter Testing Maths when Testing Maths already exists	Extra whitespace is removed/normalised and
S-04A	Subject Management	Prevent duplicate caused by excessive internal whitespace	Enter Testing Maths when Testing Maths already exists	Extra whitespace is normalised and duplicate is rejected
S-04AR	Subject Management	Retest duplicate with excessive whitespace	Enter Testing Maths when Testing Maths already exists	Duplicate is rejected
S-05	Subject Management	Prevent case-insensitive duplicate	Enter existing subject using different capitalisation. e.g.	Duplicate error shown and subject is not added
S-06	Subject Management	Remove unnecessary whitespace	Enter Arabic	Subject is stored without leading or trailing whitespace
S-07	Subject Management	Generate unique subject ID	Highest existing subject ID was SUB011; add new	New subject ID should be one greater than the highest
S-08	Subject Management	Rename subject	Rename French to Spanish	Subject name is updated successfully while keeping the
S-09	Subject Management	Prevent blank rename	Rename Spanish to spaces only	Error shown and existing name remains unchanged
S-10	Subject Management	Prevent duplicate rename	Rename Spanish to an existing subject using	Duplicate error shown and subject name remains
S-11	Subject Management	Delete unused subject	Delete Spanish, which is not assigned to any class	Subject is removed successfully

S-12	Subject Management	Prevent deletion of subject in use	Delete a subject assigned to an existing class	Deletion is blocked and subject remains
SU-01	Subject CSV Upload	Upload valid subject CSV	CSV containing new subjects such as Biologv. Chemistrv.	All valid unique subjects are added with new SUB### IDs
SU-02	Subject CSV Upload	Reject missing required column	Upload CSV with name instead of subject name	Error identifies the missing required column and nothing is added
SU-03	Subject CSV Upload	Handle blank subject rows	CSV containing blank lines between valid subject names	Blank rows are ignored and valid subjects are still added
SU-05	Subject CSV Upload	Prevent duplicate within upload	Upload CSV containing Sociology multiple times	Subject is added once only
SU-06	Subject CSV Upload	Prevent case-insensitive upload duplicate	CSV contains Drama, drama, DRAMA. DrAmA	Subject is added once only
SU-07	Subject CSV Upload	Remove upload whitespace	CSV contains a subject with leading/trailing spaces	Subject is stored without unnecessary spaces
SU-07	Subject CSV Upload	Remove upload whitespace	Subject contains leading/trailing spaces	Subject saved without unnecessary spaces
SU-08	Subject CSV Upload	Handle CSV containing only existing subjects	Upload subjects that already exist	No new subjects added and warning displayed
SU-09	Subject CSV Upload	Download correct template	Click Download Subject Template	Template downloads with subject name header
SU-10	Subject CSV Upload	Display successful upload message	Upload valid new subjects	Subjects added and success message states how many
SU-11	Subject CSV Upload	Reject CSV containing no valid subjects	Upload CSV containing only blank subject rows	No data added and error displayed
SU-12	Subject CSV Upload	Prevent duplicate with excessive internal whitespace	Upload Media Studies when Media Studies exists	Whitespace normalised and duplicate rejected
SU-13	Subject CSV Upload	Prevent duplicate with case and whitespace differences	Upload ARABIC when Arabic exists	Spaces/case ignored and duplicate rejected
SU-14	Subject CSV Upload	Handle mixed valid, duplicate and blank rows	Upload existing subjects, blank rows and duplicate	Existing/blank rows ignored and Criminology added once
SU-10R	Subject CSV Upload	Retest successful upload message after fix	Upload valid CSV containing new subjects	Success message states how many subjects were uploaded
C-01	Class Management	Add valid class	Add valid class with subject and year group	Class added with new C### ID, correct subject and year
C-02	Class Management	Prevent blank class name	Leave class name blank	Error displayed and class not added
C-03	Class Management	Prevent whitespace-only class name	Enter spaces only	Spaces removed and class rejected
C-04	Class Management	Prevent exact duplicate class	Enter exact name of existing class	Duplicate error displayed
C-05	Class Management	Prevent case-insensitive duplicate	Enter existing class with different capitalisation	Duplicate rejected regardless of case
C-06	Class Management	Prevent duplicate caused by excessive internal whitespace	Enter 7B Maths when 7B Maths exists	Extra whitespace normalised and duplicate rejected
C-07	Class Management	Remove leading/trailing whitespace	Add class with spaces before/after name	Class saved without unnecessary spaces
C-08	Class Management	Generate unique class ID	Add class after existing C### IDs	New ID is highest existing ID + 1
C-09	Class Management	Store correct subject relationship	Add class using existing subject	Correct subject_id stored in classes.csv
C-10	Class Management	Minimum year boundary	Add class with year 1	Year 1 accepted
C-11	Class Management	Maximum year boundary	Add class with year 13	Year 13 accepted
C-12	Class Management	Rename class	Rename existing class	Name updates while class ID remains unchanged
C-13	Class Management	Prevent blank update	Change class name to spaces only	Error shown and original retained
C-14	Class Management	Prevent duplicate update	Rename class to existing class name	Duplicate error shown
C-15	Class Management	Update subject	Change class to another existing subject	Subject changes while class ID stays same
C-16	Class Management	Update year group	Change year group of unused class	Year updates while class ID stays same
C-17	Class Management	Delete unused class	Delete class with no students	Class removed successfully

CU-01	Class CSV Upload	Download correct template	Download class template	Contains class name.subject name.vea
CU-02	Class CSV Upload	Upload valid classes	Upload multiple valid new classes	Valid classes added with C### IDs
CU-03	Class CSV Upload	Reject missing required column	Upload CSV missing required column	Missing-column error shown
CU-04	Class CSV Upload	Reject invalid year groups	Upload year 0, 14 or text	Invalid rows rejected
CU-05	Class CSV Upload	Mixed valid and invalid years	Upload valid and invalid rows together	Only valid rows added
CU-06	Class CSV Upload	Reject nonexistent subject	Use subject not in system	Error shown and upload stopped
CU-07	Class CSV Upload	Ignore blank rows	Valid classes with blank lines	Blank lines ignored
CU-08	Class CSV Upload	Skip existing class	Upload existing class	Existing class not duplicated
CU-09	Class CSV Upload	Prevent duplicate within same CSV	Same class repeated	Class added once
CU-10	Class CSV Upload	Prevent case-insensitive duplicate	Same class with different capitalisation	Class added once
CU-11	Class CSV Upload	Remove leading/trailing whitespace	Class contains spaces before/after name	Saved without unnecessary spaces
CU-12	Class CSV Upload	Prevent duplicate caused by internal whitespace	Upload existing class name with excessive spaces inside	Internal whitespace normalised and duplicate
CU-13	Class CSV Upload	Existing and new classes together	Upload existing classes plus one new class	Only new class added
CU-14	Class CSV Upload	Upload only existing classes	All uploaded classes already exist	No classes added and warning shown
CU-15	Class CSV Upload	Display successful upload message	Upload valid new classes	Success message states number uploaded
CI-01	Class Relationship	Prevent deletion of class containing students	Delete class that has students assigned	Deletion blocked to prevent orphan class-student links
CI-02	Class Relationship	Prevent year mismatch for assigned students	Change year of class containing students	Update blocked if assigned students belong to different
C-06	Add Class	Check that class names with excessive internal spaces are	Existing class: 7B Maths then attempt to add 7B Maths	The second class should not be added and an error should state
CU-12	Upload Class CSV	Check that uploaded class names with excessive internal	Existing class: 9Z Science Test; CSV contains 9Z Science Test	The uploaded class should be recognised as an existing class
ST-01	Student Relationship	Save student's selected class when adding a student	Add a new student and assign them to an existing class	Student is created and the selected class is saved and
ST-01R	Student Relationship	Retest saving student's selected class after fix	Add a new student and assign them to an existing class	Student is created and assigned class appears in the Classes
CI-01	Class Relationship	Prevent deletion of class containing students	Attempt to delete a class with at least one student assigned	Deletion is blocked and the class remains in the system
CI-01R	Class Relationship	Retest deletion of class containing students after fix	Attempt to delete a class with a student assigned, then test an	Class containing students is not deleted; unused class can still be
ST-01	Student Management	Add valid student	Enter valid student name, year group and select an existing	Student added with new S### ID and selected class
ST-02	Student Management	Prevent blank student name	Leave student name blank and select a class	Error displayed and student not added
ST-03	Student Management	Prevent whitespace-only student name	Enter spaces only as student name	Whitespace removed, error displayed and student not added
ST-04	Student Management	Remove unnecessary whitespace from student name	Enter student name with leading, trailing or repeated	Student name stored with normal spacing
ST-04R	Student Management	Retest student-name whitespace after fix	Enter student name containing excessive spaces	Name stored with single spaces and no unnecessary whitespace
ST-05	Student Management	Minimum year boundary	Add student in Year 1 with an available Year 1 class	Student accepted
ST-06	Student Management	Maximum year boundary	Add student in Year 13 with an available Year 13 class	Student accepted
ST-07	Student Management	Require at least one class	Enter valid student details but select no class	Error displayed and student not added
ST-08	Student Management	Handle year group with no classes	Select a year group for which no classes exist	Warning displayed and student cannot be added without a class
ST-09	Student Relationship	Save selected class when adding student	Add student and select one existing class	Student created and selected class stored and displayed

ST-09R	Student Relationship	Retest saving selected class after fix	Add new student and assign one class	Student created and assigned class displayed in Classes column
ST-10	Student Relationship	Assign student to multiple different subjects	Select classes such as Maths, English and Science for same student	Student assigned to all selected classes
ST-11	Student Relationship	Prevent two classes for same subject	Select two Maths classes for the same student	Student should not be added to both classes
ST-11R	Student Relationship	Retest same-subject class restriction	Select two classes linked to the same subject	Error displayed and student not added with conflicting classes
ST-12	Student Management	Allow students with identical names	Add two different students with the same name	Both students accepted because student_id is the unique
ST-13	Student Management	Generate unique student ID	Add student when previous S### IDs already exist	New ID should be one greater than highest existing student ID
ST-14	Student Management	Rename student	Change existing student's name	Name updated while student ID remains unchanged
ST-15	Student Management	Prevent blank student rename	Change name to spaces only	Error displayed and original name retained
ST-16	Student Management	Normalise whitespace during rename	Rename student using excessive internal/outer spaces	Updated name stored with normal spacing
ST-17	Student Relationship	Change student's assigned class	Replace current class with another class in same year	Old relationship removed and new relationship created
ST-18	Student Relationship	Add another different-subject class during update	Existing student has one class; add another subject class	Both valid classes stored
ST-19	Student Relationship	Remove one of several classes	Student belongs to multiple classes; deselect one	Deselected relationship removed while others remain
ST-20	Student Relationship	Prevent removing every class	Deselect all classes when updating student	Error displayed and student retains existing data
ST-21	Student Relationship	Prevent two classes for same subject during edit	Edit student and select two Maths classes	Update blocked
ST-21R	Student Relationship	Retest same-subject restriction during edit	Select two classes belonging to same subject	Error displayed and update blocked
ST-22	Student Relationship	Change student year and class	Change student's year group and select a class from new year	Year and class relationships both update
ST-23	Student Relationship	Change to year with no classes	Select year group with no available classes	Warning displayed and update cannot be completed
ST-24	Student Management	Delete student	Delete existing student	Student removed successfully
ST-25	Student Relationship	Delete student assigned to classes	Delete student belonging to one or more classes	Student and all class-student links removed
ST-26	Student Management	Cancel student deletion	Click Cancel in delete confirmation	Student remains unchanged
STU-01	Student CSV Upload	Download correct student template	Click Download Student Template	Template contains name, year_group, classes
STU-02	Student CSV Upload	Upload valid student	Upload valid name, year and existing class	Student and relationship added
STU-03	Student CSV Upload	Upload multiple valid students	CSV contains several valid students	All valid students added with separate S### IDs
STU-04	Student CSV Upload	Reject missing required column	Remove one required CSV column	Missing-column error displayed and upload stopped
STU-05	Student CSV Upload	Ignore blank rows	CSV contains valid rows separated by blank rows	Blank rows ignored and valid students processed
STU-06	Student CSV Upload	Reject year below minimum	Upload student with year 0	Invalid row rejected
STU-07	Student CSV Upload	Reject year above maximum	Upload student with year 14	Invalid row rejected
STU-08	Student CSV Upload	Reject non-numeric year	Set year group to abc	Invalid row rejected
STU-09	Student CSV Upload	Reject missing student name	Student name blank	Invalid row removed
STU-10	Student CSV Upload	Reject missing class	Classes field blank	Invalid row removed
STU-11	Student CSV Upload	Reject nonexistent class	Upload student with class not present in classes.csv	Upload stopped and missing class identified
STU-12	Student CSV Upload	Case-insensitive class lookup	Existing 12A English; CSV contains 12a english	Existing class recognised
STU-13	Student CSV Upload	Remove leading/trailing class whitespace	CSV class contains spaces before or after name	Existing class recognised

STU-14	Student CSV Upload	Handle excessive internal whitespace in class	Existing 12A English; upload 12A English	Whitespace normalised and existing class recognised
STU-14R	Student CSV Upload	Retest excessive class whitespace after fix	Upload 12A English where 12A English exists	Class recognised and student uploaded
STU-15	Student CSV Upload	Normalise uploaded student-name whitespace	Student name contains excessive spaces	Student stored with normal spacing
STU-16	Student CSV Upload	Assign multiple classes using semicolon	12A Maths;12A English	Student assigned to both classes
STU-17	Student CSV Upload	Handle whitespace around multiple class names	12A Maths ; 12A English	Both names normalised and recognised
STU-18	Student CSV Upload	Prevent class/student year mismatch	Year 12 student assigned to Year 11 class	Upload blocked with year-group error
STU-19	Student CSV Upload	Prevent two classes for same subject	Student uploaded with two Maths classes	Upload blocked
STU-19R	Student CSV Upload	Retest duplicate-subject validation	Student assigned to two classes whose subject_id is Maths	Upload blocked with duplicate-subject error
STU-20	Student CSV Upload	Allow multiple different subjects	Student assigned to Maths and English classes	Upload accepted
STU-21	Student CSV Upload	Allow duplicate student names	CSV contains two students with same name	Both rows accepted and given different IDs
STU-22	Student CSV Upload	Reject upload containing no valid students	CSV contains only blank/invalid rows	Error shown and nothing added
STU-23	Student CSV Upload	Display successful upload message	Upload valid students	Success message states number uploaded

assessments.py tes

AS-01	Assessment Management	Add normal valid assessment	Valid class/student, Maths Mock 1, valid date, score 15, max 20,	Assessment added successfully
AS-02	Assessment Relationship	No classes available	No classes exist	Warning displayed and assessment cannot be added
AS-03	Assessment Relationship	Class linked to nonexistent subject	Class has invalid subject relationship	Error displayed and assessment cannot be added
AS-04	Assessment Relationship	Class contains no students	Select class with no assigned students	Warning displayed and assessment cannot be added
AS-05	Assessment Relationship	Only display students belonging to selected class	Select a class containing known students	Only students assigned to that class shown
AS-06	Assessment Management	Prevent blank assessment title	Leave title blank	Error displayed and assessment not added
AS-07	Assessment Management	Prevent whitespace-only title	Enter spaces only	Whitespace removed and title rejected
AS-08	Whitespace	Remove leading/trailing title whitespace	Enter Maths Mock 1	Store as Maths Mock 1
AS-09	Whitespace	Normalise internal assessment-title whitespace	Enter title containing excessive internal spaces	Repeated spaces should be reduced to single spaces
AS-09R	Whitespace	Retest internal title whitespace	Enter title with excessive internal spaces after fix	Title stored with normal spacing
AS-10	Score	Minimum maximum score	Maximum score 1	Assessment accepted
AS-11	Score	Score equal to zero	Score 0, max 20	Assessment accepted
AS-12	Score	Score equal to maximum	Score 20, max 20	Assessment accepted
AS-13	Pass Mark	Pass mark equal to zero	Pass mark 0	Assessment accepted
AS-14	Pass Mark	Pass mark equal to maximum	Pass mark 20, max 20	Assessment accepted
AS-15	Score	Score below pass mark	Score 5, max 20, pass 10	Valid assessment representing a failed result
AS-16	Date	Prevent future date	Attempt to select future assessment date	Future date cannot be saved
AS-17	Duplicate	Prevent exact duplicate assessment	Same student, subject, title and date twice	Second assessment rejected
AS-18	Duplicate	Prevent case-insensitive duplicate title	Maths Mock 1 followed by MATHS MOCK 1	Duplicate rejected
AS-19	Duplicate	Prevent duplicate caused by internal whitespace	Existing Maths Mock 1; add Maths Mock 1	Duplicate should be rejected

AS-19R	Duplicate	Retest whitespace duplicate after fix	Existing Maths Mock 1; add whitespace variation	Duplicate rejected
AS-20	Duplicate	Same title on different date	Same student/title but different date	Separate assessment allowed
AS-21	Duplicate	Same assessment title for different subject	Same student/title/date but different subjects	Both assessments allowed
AS-22	Duplicate	Same assessment for different students	Same title/subject/date for different student IDs	Both records allowed
AS-23	Success Message	Display successful add message	Add valid assessment	Success message displayed

assessment csv upload

ASU-01	Assessment CSV Upload	Download correct template	Download Assessment Template	Seven required headers present
ASU-02	Assessment CSV Upload	Upload valid assessment	Complete valid CSV row	Assessment uploaded successfully
ASU-03	Assessment CSV Upload	Upload multiple valid assessments	Several valid rows	All rows added
ASU-04	Assessment CSV Upload	Reject missing required column	Remove required header	Error identifies missing column
ASU-05	Assessment CSV Upload	Reject empty CSV	Headers only	Error displayed
ASU-06	Assessment CSV Upload	Reject missing required value	Leave required cell blank	Upload rejected
ASU-07	Whitespace	Remove outer student-ID whitespace	S001	Recognised as S001
ASU-08	Student ID	Reject internal student-ID whitespace	S 001	Student ID not recognised
ASU-09	Whitespace	Remove outer class-name whitespace	12A English	Class recognised
ASU-10	Whitespace	Normalise internal class-name whitespace	Existing 12A English; upload 12A English	Existing class recognised
ASU-10R	Whitespace	Retest internal class whitespace	Upload 12A English after fix	Recognised as 12A English
ASU-11	Whitespace	Remove outer assessment-title whitespace	English Mock 1	Stored without outer spaces
ASU-12	Whitespace	Normalise internal assessment-title whitespace	English Mock 1	Stored as English Mock 1
ASU-12R	Whitespace	Retest uploaded title whitespace	Upload excessive internal title spaces	Stored with normal spacing
ASU-13	Student	Reject nonexistent student	Student ID S999	Upload rejected
ASU-14	Class	Reject nonexistent class	Class 99Z English	Upload rejected
ASU-15	Class	Case-insensitive class lookup	Existing 12A English; CSV 12a english	Class recognised
ASU-16	Relationship	Student not assigned to class	Valid student ID with unrelated class	Upload rejected
ASU-17	Relationship	Class has invalid subject relationship	Class references nonexistent subject	Upload rejected
ASU-18	Numeric	Reject non-numeric score	Score abc	Upload rejected
ASU-19	Numeric	Reject non-numeric maximum score	Maximum xyz	Upload rejected
ASU-20	Numeric	Reject non-numeric pass mark	Pass mark text	Upload rejected
ASU-21	Max Score	Reject maximum score zero	Max 0	Upload rejected
ASU-22	Max Score	Reject negative maximum score	Max -20	Upload rejected
ASU-23	Score	Reject negative score	Score -1	Upload rejected
ASU-24	Score	Reject score above maximum	Score 25, max 20	Upload rejected
ASU-25	Pass Mark	Reject negative pass mark	Pass -1	Upload rejected

ASU-26	Pass Mark	Reject pass mark above maximum	Pass 25, max 20	Upload rejected
ASU-27	Numeric	Accept decimal assessment values	Score 7.5, max 10, pass 5.5	Valid assessment accepted
ASU-28	Date	Reject invalid date	not-a-date	Upload rejected
ASU-29	Date	Validate assessment date input	Invalid/non-supported date input	Invalid date rejected
ASU-30	Date	Reject future assessment date	Date after current day	Upload rejected
ASU-31	Date	Accept old valid date	15/01/2020	Assessment accepted
ASU-32	Duplicate	Prevent existing exact duplicate	Same student/subject/title/date already stored	Upload rejected
ASU-33	Duplicate	Case-insensitive CSV duplicate	Existing Mock 1; upload MOCK 1	Duplicate rejected
ASU-34	Duplicate	Prevent duplicate with internal title whitespace	Existing Maths Mock 1; CSV Maths Mock 1	Duplicate rejected
ASU-34R	Duplicate	Retest whitespace duplicate	Upload whitespace variation after fix	Duplicate rejected
ASU-35	Duplicate	Prevent duplicate within same CSV	Same assessment appears twice	Whole upload rejected
ASU-36	Duplicate	Same title on different dates	Same student/title, different dates	Both records allowed
ASU-37	Duplicate	Same assessment for different students	Different IDs, same title/date	Both records allowed
ASU-38	Mixed Upload	Existing duplicate plus valid new assessment	One duplicate and one new record	Whole upload stopped
ASU-39	Mixed Upload	Invalid student plus valid assessment	S999 plus valid row	Whole upload stopped
ASU-40	CSV Structure	Ignore additional irrelevant column	Required fields plus Notes	Valid data imported
ASU-41	Success Message	Display correct upload count	Upload 3 valid assessments	Message states 3 uploaded
ASU-42	Subject Relationship	Derive subject from selected class	Upload valid class/student assessment	Correct subject saved automatically

edit and delete assess

ASE-01	Assessment Management	Select existing assessment	Click assessment table row	Correct management section displayed
ASE-02	Assessment Management	Prefill existing values	Select known assessment	Existing title/date/score/max/pass
ASE-03	Assessment Management	Rename assessment	Change to valid new title	Assessment renamed
ASE-04	Assessment Management	Prevent blank rename	Rename to blank/spaces	Error shown and original retained
ASE-05	Whitespace	Remove outer whitespace during rename	Mock Exam	Store Mock Exam
ASE-06	Whitespace	Normalise internal whitespace during rename	Mock Exam	Store Mock Exam
ASE-06R	Whitespace	Retest internal rename whitespace	Rename using excessive spaces	Store normally-spaced title
ASE-07	Score	Update student score	Change valid score	New score saved
ASE-08	Score	Increase maximum score	Change max to higher valid value	New maximum saved
ASE-09	Score Integrity	Lower maximum below student's existing score	Existing max 100, score 75; change max to 50	Update should be rejected because 75 exceeds new
ASE-09R	Score Integrity	Retest maximum below student score	Max 100/score 75, change max to 50	Error shown and original score retained
ASE-10	Pass Mark	Lower maximum below previous pass mark	Existing pass mark exceeds newly-selected maximum	Pass mark automatically adjusts to new maximum
ASE-11	Date	Change assessment date	Select valid past/current date	New date saved
ASE-12	Date	Prevent future date during edit	Attempt future date	Future date unavailable

ASE-13	Duplicate	Update assessment into existing duplicate	Make title/date match another record	Update rejected
ASE-14	Duplicate	Update record without changing identifying values	Press Update without changing title/date	Record should not identify itself as duplicate
ASE-15	Delete	Delete assessment	Confirm deletion	Assessment removed
ASE-16	Delete	Cancel assessment deletion	Click Cancel	Assessment retained
ASE-17	Relationship	Display assessment whose student no longer exists	Assessment references deleted student	Page remains usable and missing student identified
Test ID	Category	Test being carried out	Test input / condition	Expected result
ASI-01	Data Integrity	Delete student with existing assessment records	Give John 2020 assessment records, then delete John from	Student deletion should be blocked while assessment
attendance uploa				
AT-01	Attendance Management	Add normal valid attendance	Existing student, total 100, attended 96	Attendance record added successfully
AT-02	Attendance Management	Minimum total sessions	Total 1, attended 1	Valid record accepted
AT-03	Attendance Management	Zero sessions attended	Total 20, attended 0	Valid record accepted
AT-04	Attendance Management	All sessions attended	Total 20, attended 20	Valid record accepted
AT-05	Attendance Management	Normal partial attendance	Total 20, attended 15	Valid record accepted
AT-06	Attendance Boundary	Prevent zero total sessions	Attempt total 0	UI prevents total below 1
AT-07	Attendance Boundary	Prevent negative attended sessions	Attempt attended below 0	UI prevents negative value
AT-08	Attendance Boundary	Prevent attended greater than total during Add	Total 20, attempt attended 21	UI prevents value above total
AT-09	Attendance Relationship	Handle no classes existing	No classes in system	Warning displayed and attendance cannot be added
AT-10	Attendance Relationship	Handle class containing no students	Select empty class	Warning displayed and attendance cannot be added
AT-11	Attendance Relationship	Only show students belonging to selected class	Select class with known students	Only assigned students shown
AT-12	Attendance Relationship	Student belongs to multiple classes	Select either class containing same student	Student available through either assigned class
AT-13	Attendance Duplicate	Prevent second attendance record	Student already has attendance	Error shown and duplicate not added
AT-14	Attendance Duplicate	Prevent duplicate through another class	Student has attendance and belongs to multiple classes	Second record still rejected
AT-15	Success Message	Display successful add message	Add valid attendance	Success message appears
AT-16	Storage	Save correct student ID	Add attendance for known S#### student	Correct ID stored
AT-17	Storage	Save correct attendance values	Total 100, attended 96	Values stored unchanged
ATD-01	Attendance Display	Display attendance table	Attendance records exist	Records shown in Current Attendance
ATD-02	Attendance Display	Display correct student name	Select normal existing student attendance	Correct student name shown
ATD-02R	Attendance Display	Retest student name after fix	Select attendance belonging to existing student	Actual student's name shown
ATD-03	Attendance Calculation	Calculate 96% attendance	Total 100, attended 96	96% displayed
ATD-04	Attendance Calculation	Calculate 75% attendance	Total 20, attended 15	75% displayed
ATD-05	Attendance Calculation	Calculate 100% attendance	Total 20, attended 20	100% displayed
ATD-06	Attendance Calculation	Calculate 0% attendance	Total 20, attended 0	0% displayed
ATD-07	Attendance Display	Handle attendance belonging to deleted student	Attendance remains for deleted student	Page remains usable and missing student identified

ATD-08	Empty State	Display no-attendance message	attendance.csv contains no records	Information message displayed
ATE-01	Attendance Management	Select existing attendance row	Click attendance row	Manage Selected Attendance opens
ATE-01R	Attendance Management	Retest attendance row selection	Click attendance row after fix	Management section opens without error
ATE-02	Attendance Management	Existing total prefilled	Select record with total 100	Total Sessions shows 100
ATE-03	Attendance Management	Existing attended value prefilled	Select record attended 96	Sessions Attended shows 96
ATE-04	Attendance Update	Increase total sessions	Change 100 to 110	Valid update accepted
ATE-05	Attendance Update	Increase sessions attended	Total 110, attended 105	Valid update accepted
ATE-06	Attendance Update	Decrease sessions attended	96 to 90	Valid update accepted
ATE-07	Attendance Update	Set attended to zero	Total 100, attended 0	Valid update
ATE-08	Attendance Update	Set attended equal to total	Total 100, attended 100	Valid update
ATE-09	Attendance Integrity	Reduce total below existing attended value	Existing 100/96; change total to 90	Invalid update should be rejected
ATE-09R	Attendance Integrity	Retest reducing total below attended value	Existing 100/96; change total to 90	Error displayed and values not silently altered
ATE-10	Attendance Boundary	Prevent attended greater than total during edit	Total 90, attended 96	Error displayed and update rejected
ATE-11	Attendance Calculation	Recalculate absences	Total 100, attended 94	Absences displays 6
ATE-12	Attendance Calculation	Recalculate attendance percentage	Total 100, attended 94	94.0% displayed
ATE-13	Attendance Update	Save valid attendance edit	Change to valid values	CSV updates and success message displayed
ATE-14	Attendance Update	Student ID remains unchanged	Edit attendance numbers	Same student ID retained
ATE-15	Attendance Update	Reopen updated record	Save edit then select record again	Updated values persist
ATX-01	Attendance Delete	Delete attendance record	Select and confirm delete	Attendance removed
ATX-02	Attendance Delete	Cancel deletion	Click Cancel	Attendance remains
ATX-03	Data Integrity	Attendance deletion does not delete student	Delete student's attendance	Student remains
ATX-04	Data Integrity	Attendance deletion does not remove class membership	Delete attendance	Student remains assigned to classes
ATX-05	Attendance Delete	Re-add attendance after deleting it	Delete then add new attendance for student	New record can be added
ATX-06	Success Message	Display deletion success message	Confirm attendance deletion	Success message shown
ATU-01	Attendance CSV Upload	Download correct template	Click Download Attendance Template	Correct three headers downloaded
ATU-02	Attendance CSV Upload	Upload one valid record	Valid S###, total and attended	Record uploaded
ATU-03	Attendance CSV Upload	Upload multiple valid records	Several valid students	All records added
ATU-04	CSV Structure	Missing student_id column	Remove student_id	Missing-column error
ATU-05	CSV Structure	Missing total_sessions column	Remove total_sessions	Missing-column error
ATU-06	CSV Structure	Missing sessions_attended column	Remove sessions_attended	Missing-column error
ATU-07	CSV Structure	Empty CSV	Headers only	Error states no attendance records
ATU-08	Missing Data	Missing student ID	Blank student ID	Upload rejected
ATU-09	Missing Data	Missing total sessions	Blank total	Upload rejected

ATU-10	Missing Data	Missing attended sessions	Blank attended	Upload rejected
ATU-11	Missing Data	Multiple missing values	Several required cells blank	Upload rejected
ATU-12	Whitespace	Remove outer student-ID whitespace	S001	Treated as S001
ATU-13	Student ID	Reject internal ID whitespace	S 001	Not treated as S001
ATU-14	Student ID	Different case student ID	s001 instead of S001	Invalid ID not silently changed
ATU-15	Student	Reject nonexistent student	S999	Upload stopped
ATU-16	Numeric	Reject text total sessions	abc	Upload rejected
ATU-17	Numeric	Reject text sessions attended	xyz	Upload rejected
ATU-18	Numeric	Text in both numeric fields	abc / xyz	Upload rejected
ATU-19	Whole Number	Reject decimal total sessions		10.5 Upload rejected
ATU-20	Whole Number	Reject decimal sessions attended		5.5 Upload rejected
ATU-21	Whole Number	Reject decimals in both fields	10.5 / 5.5	Upload rejected
ATU-22	Total Sessions	Reject total equal to zero	0 / 0	Upload rejected
ATU-23	Total Sessions	Reject negative total		-2 Upload rejected
ATU-24	Sessions Attended	Reject negative attended	10 / -1	Upload rejected
ATU-25	Sessions Attended	Reject attended above total		10-Dec Upload rejected
ATU-26	Boundary	Attended equal to total		10-Oct Valid record
ATU-27	Boundary	Attended equal to zero	10 / 0	Valid record
ATU-28	Boundary	Smallest valid attendance record	1 / 0	Valid record
ATU-29	Duplicate	Same student twice in one CSV	S001 listed twice	Whole upload stopped
ATU-30	Duplicate	Same student repeated three times	S001 three times	Duplicate student identified
ATU-31	Duplicate	Multiple duplicated students	S001 twice and S002 twice	Both IDs identified
ATU-32	Existing Attendance	Upload attendance that already exists	S001 already has attendance	Upload rejected
ATU-33	Mixed Upload	Existing attendance plus valid new student	S001 existing, S002 new	Whole upload stopped
ATU-34	Mixed Upload	Invalid student plus valid student	S999 plus valid S002	Whole upload stopped
ATU-35	Mixed Upload	Invalid numeric row plus valid row	One invalid and one valid row	Whole upload stopped
ATU-36	CSV Structure	Extra irrelevant column	Required columns plus Notes	Extra column ignored
ATU-37	Success Message	Successful single CSV upload	Upload one valid record	Message states 1 record uploaded
ATU-38	Success Message	Successful multiple CSV upload	Upload 3 records	Message states 3 records uploaded
ATU-39	Storage	Store session values as integers	Upload 100 / 96	Whole-number values saved
ATU-40	Duplicate	Upload same student after previous successful upload	Student already added via earlier upload	Existing-attendance error
ATI-01	Data Integrity	Delete student with existing attendance	Add attendance for student then delete student from Students	Student deletion should be blocked while attendance
ATI-02	Data Integrity	Remove student from one class while retaining another	Student has attendance and belongs to multiple classes	Overall attendance remains associated with student

ATI-03	Data Integrity	Change student's class	Move student to another valid class	Attendance stays associated with same student ID
ATI-04	Data Integrity	Change student's year/class	Move student to new valid year/class	Existing overall attendance remains associated with student
ATI-05	Attendance Relationship	Student displayed under current assigned class	Change student's class then open Add Attendance	Student appears under currently assigned class

Actual result	Pass / Fail	Action / Fix
Record identified as valid	Pass	No change required
Record identified as valid	Pass	No change required
Record identified as valid	Pass	No change required. A failed assessment is still valid input data
Record identified as valid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Converted to missing value and identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Converted to missing value and identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as valid	Pass	No change required
Record identified as valid	Pass	No change required
Converted to missing value and identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as valid	Pass	No change required
Record identified as valid	Pass	No change required
Date converted to missing value and record identified	Pass	No change required
Record identified as invalid	Pass	No change required
Record incorrectly identified as valid	Fail	Future dates were not being checked. Added validation to ensure assessment date is not later than the current date
Record identified as invalid	Pass	Future-date validation successfully fixed the issue
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Invalid numeric values converted to missing values	Pass	No change required
Record identified as invalid	Pass	No change required
NULL values treated as missing and record	Pass	No change required
Record identified as valid	Pass	No change required

Record identified as valid	Pass	No change required
Record identified as valid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Converted to missing value and identified as invalid	Pass	No change required
Converted to missing value and identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
Record identified as invalid	Pass	No change required
NULL treated as missing and record identified as invalid	Pass	No change required
NULL treated as missing and record identified as invalid	Pass	No change required
Average 13.0, change +2.0, previous failure 0	Pass	No change required
Average 8.5, change -1.0, previous failure 1	Pass	No change required
<NA> values and Insufficient data returned	Pass	No change required
4 absences	Pass	No change required
96.00%	Pass	No change required
All calculated values matched expected values	Pass	No change required
Refactoring		
Existing tests continued to run correctly	Pass	Refactoring kept
Same assessment outputs produced and validation	Pass	Duplicated validation code removed
All tests passed	Pass	No further change required
All 15 tests passed	Pass	No further change required
Record marked Ready	Pass	No change required
Record marked Insufficient data	Pass	No change required
Record retained and marked Insufficient data	Pass	No change required
Student retained with missing features and Maths and English retained separately, both with 3	Pass	No change required
1		

T001 retained with all four features and status Ready	Pass	No change required
T002 retained and marked Insufficient data	Pass	No change required
T003 retained with missing absences and marked	Pass	No change required
T004 retained with missing features and marked	Pass	No change required
Maths and English retained separately and both	Pass	No change required
S001 predicted Not At Risk	Pass	No change required
S002 predicted At Risk	Pass	No change required
S003 predicted Not At Risk	Pass	No change required
S004 predicted Not At Risk	Pass	No change required
S005 predicted Not At Risk	Pass	No change required
S006 predicted Not At Risk	Pass	No change required
S007 remained Insufficient data	Pass	No change required
S008 retained and remained Insufficient data	Pass	No change required
S009 retained and remained Insufficient data	Pass	No change required
S010 retained and remained Insufficient data	Pass	No change required
10/10 status comparisons retained	Pass	No change required
10/10 prediction comparison	Pass	No change required

Tests

Testing Maths was added successfully. appeared	Pass	No action required
Error message displayed stating Subject name is	Pass	No action required
Spaces were removed and Subject name is	Pass	No action required
This subject already exists. was displayed	Pass	No action required
Subject was added because internal	Fail	Normalise repeated whitespace before checking/saving subject names
Testing Maths was added as a separate subject	Fail	Normalise repeated spaces in subject names before duplicate checking and saving
Subject was initially added as a separate record	Fail	Normalised both entered and existing subject names before duplicate comparison
This subject already exists. displayed and no duplicate	Pass	Fix successful
All capitalisation variations displayed This	Pass	No action required
Subject was saved as Arabic in the table and	Pass	No action required
French was added with ID SUB012	Pass	No action required
French was renamed to Spanish. ID remained	Pass	No action required
Subject name is required. was displayed.	Pass	No action required
All variations were rejected with This	Pass	No action required
Spanish was deleted from the table and CSV	Pass	No action required

Error displayed: This subject cannot be deleted	Pass	No action required
All new subjects were added successfully and a	Pass	No action required
Error displayed: Missing required columns:	Pass	No action required
Blank lines were ignored and valid subjects were	Pass	No action required
Only one Sociology record was added	Pass	No action required
Only one Drama record was added	Pass	No action required
Subject was saved normally with no leading	Pass	No action required
Subject was saved normally with no leading or	Pass	No action required
No duplicates were added and warning displayed	Pass	No action required
Template downloaded correctly with the required	Pass	No action required
Subjects were added correctly but no success	Fail	Session-state key used after upload does not match key checked at top of page
No valid subjects were found in the uploaded file.	Pass	No action required
Existing Media Studies was recognised and no	Pass	No action required
Existing subject recognised and duplicate was not	Pass	No action required
Only one Criminology was added, but no upload	Fail	Same session-state success-message key mismatch as SU-10
Success message displayed correctly after upload	Pass	Changed the session-state key so the saved upload message matches the key checked after rerun
Class added successfully with correct ID, subject	Pass	No action required
Blank class was rejected correctly	Pass	No action required
Whitespace-only class was rejected	Pass	No action required
Exact duplicate was rejected	Pass	No action required
Different capitalisation versions were rejected	Pass	No action required
Class was accepted as a separate class because	Fail	Normalise internal whitespace before duplicate checking and saving
Leading/trailing whitespace was removed	Pass	No action required
Correct next class ID generated	Pass	No action required
Correct subject ID was stored	Pass	No action required
Class created successfully with year 1	Pass	No action required
Class created successfully with year 13	Pass	No action required
Class renamed successfully and ID	Pass	No action required
Blank update was rejected correctly	Pass	No action required
Duplicate update was rejected	Pass	No action required
Subject updated correctly	Pass	No action required
Year group updated correctly	Pass	No action required
Unused class deleted successfully	Pass	No action required

Template downloaded with correct headers	Pass	No action required	
Valid classes uploaded successfully	Pass	No action required	
Missing column correctly identified and upload	Pass	No action required	
Invalid year groups were not added	Pass	No action required	
Only valid class records were added	Pass	No action required	
Nonexistent subject correctly rejected	Pass	No action required	
Blank rows ignored and valid classes added	Pass	No action required	
Existing class was skipped	Pass	No action required	
Only one copy added	Pass	No action required	
Case variations treated as duplicates	Pass	No action required	
Leading/trailing spaces removed correctly	Pass	No action required	
Class with excessive internal whitespace could not be added	Fail	Apply same internal-whitespace normalisation to CSV class names	
Existing classes skipped and new class added	Pass	No action required	
No duplicate classes added and warning displayed	Pass	No action required	
Correct success message displayed	Pass	No action required	
Class was still allowed to be deleted even though it contained students	Fail	Check class_students.csv before deletion and block deletion when students are assigned	
Class year could still be changed while students were added	Fail	Check assigned students' year groups before allowing class year change	
Both class names were initially treated as different	Fail	Class names were normalised before validation using ".join(class_name.split())". Existing class names are also normalised using ".str.split().str.join(" ")", while existing class names are not.	
The class was initially treated as different because student was created, but the selected class was not	Fail	st.rerun() was being called before the student-class relationship was saved. Removed the premature rerun so	Pass
Student was created successfully and the class was deleted even though students were still in the class	Pass	Fix successful. Student-class relationships are now saved before st.rerun()	
Class containing a student was correctly protected	Fail	classes.py did not check class_students.csv before deletion. Added a check for the selected class_id and blocked deletion.	
Student was added successfully with a unique ID	Pass	Fix successful. Class deletion now checks student-class relationships before allowing deletion	
Student name is required.	Pass	No action required	
Whitespace was removed and blank name was allowed	Pass	No action required	
Excessive whitespace was initially not fully normalised	Fail	Changed student-name cleaning to ".join(student_name.split())"	
Student name was stored correctly after whitespace	Pass	Fix successful	
Year 1 student added correctly	Pass	No action required	
Year 13 student added correctly	Pass	No action required	
Please select at least one class. displayed	Pass	No action required	
Warning displayed stating no classes exist for the selected year	Pass	No action required	
Student was initially created but class relationship was not saved	Fail	st.rerun() was executing before class_students.csv was updated. Removed premature rerun	

Assigned class was saved and displayed correctly	Pass	Fix successful
Multiple different-subject classes were stored and System initially allowed a student to be assigned to	Pass	No action required
A student cannot be assigned to more than one	Fail	Added subject ID comparison across selected classes
Both students were created with different S#### IDs	Pass	Fix successful
Correct next S#### ID generated	Pass	No action required. Duplicate names are valid
Student renamed successfully and ID remained	Pass	No action required
Blank update was rejected and existing name remained	Pass	No action required
Updated student name was cleaned correctly	Pass	Whitespace normalisation added to Update Student
Class assignment changed correctly	Pass	No action required
Additional class was added correctly	Pass	No action required
Selected class was removed and remaining classes	Pass	No action required
Please select at least one class. displayed and invalid	Pass	No action required
Same-subject duplicate was initially possible through	Fail	Added same subject-ID duplicate validation to Update Student
Conflicting class selection was correctly rejected	Pass	Fix successful
Student year and selected class were updated correctly	Pass	No action required
Warning displayed and no class could be selected	Pass	No action required
Student deleted from students.csv and success	Pass	No action required
Student and associated class relationships were removed	Pass	No action required
Student was not deleted	Pass	No action required
Correct headers downloaded	Pass	No action required
Student uploaded successfully	Pass	No action required
Multiple students uploaded successfully	Pass	No action required
Missing required column was correctly identified	Pass	No action required
Blank rows were ignored	Pass	No action required
Year 0 student was not added	Pass	No action required
Year 14 student was not added	Pass	No action required
Non-numeric value converted to invalid/missing	Pass	No action required
Blank-name student was not uploaded	Pass	No action required
Student without a class was not uploaded	Pass	No action required
Missing class error displayed correctly	Pass	No action required
Class was recognised regardless of capitalisation	Pass	No action required
Leading/trailing whitespace handled correctly	Pass	No action required

Class was initially reported as nonexistent because only Extra spaces were removed	Fail	Added per-class normalisation using " <code>".join(class_name.split())</code>
and 12A English was Name was cleaned successfully	Pass	Fix successful
Both relationships created correctly	Pass	Added <code>.str.split().str.join(" ")</code> to uploaded names
Both class names were cleaned and recognised	Pass	No action required
Mismatched class assignment was correctly	Pass	Whitespace cleaning retained
CSV initially had no check preventing two classes for Student was correctly	Fail	No action required
prevented from joining two Student uploaded and linked	Pass	Added duplicate-subject validation using each class's <code>subject_id</code>
to both different-subject Both students were	Pass	Fix successful
successfully uploaded with No valid students were	Pass	No action required
found in the uploaded file. Correct number of uploaded students displayed	Pass	No action required. Student names are not unique identifiers
ting		
Assessment added and saved correctly	Pass	No action required
Warning displayed correctly	Pass	No action required
Invalid subject relationship detected	Pass	No action required
No-students warning displayed	Pass	No action required
Correct students displayed	Pass	No action required
Blank title rejected	Pass	No action required
Whitespace-only title rejected	Pass	No action required
Leading/trailing whitespace removed correctly	Pass	No action required
Internal whitespace was not initially normalised	Fail	Changed title cleaning to " <code>".join(assessment_title.split())</code>
Internal whitespace normalised correctly	Pass	Fix successful
Assessment accepted	Pass	No action required
Assessment accepted	Pass	No action required
Assessment accepted	Pass	No action required
Assessment accepted	Pass	No action required
Assessment accepted	Pass	No action required
Assessment accepted correctly	Pass	No action required
Future date was rejected and valid current date	Pass	No action required
Duplicate correctly rejected	Pass	No action required
Duplicate correctly recognised regardless of Assessment was initially added as a separate record	Pass	No action required
	Fail	Normalised both entered and existing assessment titles before duplicate comparison

Whitespace variation correctly recognised as	Pass	Fix successful
Assessment added successfully	Pass	No action required
Both records accepted	Pass	No action required
Both assessments accepted	Pass	No action required
Assessment added successfully message	Pass	No action required
oad		
Correct template downloaded	Pass	No action required
Assessment uploaded correctly	Pass	No action required
All assessments uploaded	Pass	No action required
Missing column correctly detected	Pass	No action required
No assessment records error displayed	Pass	No action required
Missing-value error displayed	Pass	No action required
Outer whitespace removed correctly	Pass	No action required
ID correctly treated as nonexistent	Pass	No action required
Class recognised correctly	Pass	No action required
Class initially reported as nonexistent	Fail	Added <code>.str.split().str.join(" ")</code> to uploaded class names
Class recognised correctly	Pass	Fix successful
Outer whitespace removed	Pass	No action required
Internal whitespace was initially retained	Fail	Added internal whitespace normalisation to uploaded assessment titles
Title normalised correctly	Pass	Fix successful
Missing student correctly identified	Pass	No action required
Missing class correctly identified	Pass	No action required
12a english correctly matched 12A English	Pass	No action required
Student/class mismatch correctly detected	Pass	No action required
Invalid subject link correctly detected	Pass	No action required
Numeric validation error displayed	Pass	No action required
Numeric validation error displayed	Pass	No action required
Numeric validation error displayed	Pass	No action required
Invalid maximum rejected	Pass	No action required
Invalid maximum rejected	Pass	No action required
Negative score rejected	Pass	No action required
Score above maximum rejected	Pass	No action required
Negative pass mark rejected	Pass	No action required

Invalid pass mark rejected	Pass	No action required
Decimal values accepted	Pass	No action required
Invalid date correctly rejected	Pass	No action required
Date validation behaved correctly	Pass	No action required
Future date rejected	Pass	No action required
Old valid date accepted	Pass	No action required
Existing duplicate identified	Pass	No action required
Case-insensitive duplicate identified	Pass	No action required
Initially treated as a different assessment	Fail	Normalised uploaded and existing assessment titles before duplicate checking
Duplicate correctly identified	Pass	Fix successful
Duplicate within upload correctly detected	Pass	No action required
Both accepted	Pass	No action required
Both assessments accepted	Pass	No action required
Upload stopped without partial addition	Pass	No action required
Invalid student stopped upload	Pass	No action required
Extra column ignored	Pass	No action required
Correct upload count displayed	Pass	No action required
Correct subject linked and stored	Pass	No action required
ments		
Selected assessment displayed correctly	Pass	No action required
Correct values shown	Pass	No action required
Title updated successfully	Pass	No action required
Blank rename rejected	Pass	No action required
Outer whitespace removed	Pass	No action required
Internal spaces were initially not normalised	Fail	Changed update cleaning to " <code>".join(new_assessment_title.split())</code> "
Internal whitespace correctly normalised	Pass	Fix successful
Score updated correctly	Pass	No action required
Maximum updated correctly	Pass	No action required
Score was silently changed from 75 to 50 and update	Fail	Removed automatic score clamping and added explicit <code>new_score > new_max_score</code> validation
Student score cannot be greater than maximum	Pass	Fix successful
Pass mark adjusted correctly	Pass	Intended behaviour retained
Date updated correctly	Pass	No action required
Future date prevented	Pass	No action required

Duplicate update correctly rejected	Pass	No action required
Existing record updated without false duplicate error	Pass	No action required
Assessment deleted successfully	Pass	No action required
Assessment remained unchanged	Pass	No action required
ID remained visible; name blank in table and selection	Pass	Defensive missing-student handling works correctly
Actual result	Pass / Fail	Action / Fix
Student was deleted while assessment records	Fail	Add dependency checking to Student deletion so students with assessment records cannot be deleted until those records are
id		
Attendance was added and saved correctly	Pass	No action required
Record accepted successfully	Pass	No action required
Record accepted successfully	Pass	No action required
Record accepted successfully	Pass	No action required
Record accepted successfully	Pass	No action required
Record accepted successfully	Pass	No action required
Total Sessions could not be set below 1	Pass	No action required
Negative value could not be entered	Pass	No action required
Sessions Attended was limited to maximum total	Pass	No action required
Warning displayed correctly	Pass	No action required
No-students warning displayed	Pass	No action required
Correct students appeared	Pass	No action required
Student appeared correctly in assigned classes	Pass	No action required
Existing attendance correctly detected	Pass	No action required
Duplicate rejected regardless of selected class	Pass	No action required
Attendance added successfully message	Pass	No action required
Correct student ID saved in attendance.csv	Pass	No action required
100 and 96 stored correctly	Pass	No action required
Records displayed correctly	Pass	No action required
Every selected student initially displayed	Fail	An incorrect else changed valid student names to the missing-student message
Student name displayed correctly after removing	Pass	Fix successful
Percentage calculated correctly	Pass	No action required
Percentage calculated correctly	Pass	No action required
Percentage calculated correctly	Pass	No action required
Percentage calculated correctly	Pass	No action required
Name was blank in table and selection showed Student	Pass	Defensive missing-student handling works

No attendance data has been added yet displayed	Pass	No action required
Selecting row caused TypeError: string indices	Fail	selected_student and selected_attendance were incorrectly storing individual strings instead of the selected row
Row selection worked and management controls	Pass	Used attendance_display.iloc[selected_row] to store complete selected row
Existing total displayed correctly	Pass	No action required
Existing value displayed correctly	Pass	No action required
Total updated successfully	Pass	No action required
Attendance updated correctly	Pass	No action required
Attendance updated correctly	Pass	No action required
Value accepted and saved	Pass	No action required
Value accepted and saved	Pass	No action required
Sessions attended automatically changed from	Fail	min() and max_value were automatically clamping attended value
Attended remained 96 and update was blocked because	Pass	Removed automatic clamping and added explicit validation before saving
The number of sessions attended cannot exceed the	Pass	Validation now performed when Update Attendance is clicked
Correct value displayed	Pass	No action required
Correct percentage displayed	Pass	No action required
Updated values saved successfully	Pass	No action required
Student ID remained unchanged	Pass	No action required
Correct updated values shown	Pass	No action required
Record deleted successfully	Pass	No action required
Record remained unchanged	Pass	No action required
Student remained in Students section	Pass	No action required
Class relationships remained unchanged	Pass	No action required
Attendance could be created again	Pass	No action required
Attendance deleted successfully message	Pass	No action required
Template contained student_id, total_sessions	Pass	No action required
Valid attendance uploaded successfully	Pass	No action required
Multiple attendance records uploaded	Pass	No action required
Missing column correctly identified	Pass	No action required
Missing column correctly identified	Pass	No action required
Missing column correctly identified	Pass	No action required
Correct error displayed	Pass	Error wording also corrected from assessment records to attendance records
Missing required value detected	Pass	No action required
Missing required value detected	Pass	No action required

Missing required value detected	Pass	No action required
Upload stopped correctly	Pass	No action required
Outer whitespace removed and student recognised	Pass	No action required
Student ID not recognised	Pass	Correct behaviour; internal ID whitespace should not be removed
ID treated as nonexistent	Pass	Student IDs are unique exact identifiers
Missing student correctly identified	Pass	No action required
Numeric error displayed	Pass	No action required
Numeric error displayed	Pass	No action required
Numeric validation stopped upload	Pass	No action required
Whole-number validation displayed	Pass	No action required
Whole-number validation displayed	Pass	No action required
Upload correctly stopped	Pass	No action required
Total must be greater than 0 error displayed	Pass	No action required
Invalid total correctly rejected	Pass	No action required
Negative attendance correctly rejected	Pass	No action required
Invalid record rejected	Pass	No action required
Record accepted	Pass	No action required
Record accepted	Pass	No action required
Record accepted	Pass	No action required
Duplicate ID correctly detected	Pass	No action required
Student correctly identified as duplicated	Pass	No action required
All duplicate student IDs identified	Pass	No action required
Existing attendance correctly detected	Pass	No action required
Upload stopped without partially saving S002	Pass	No action required
Missing student stopped complete upload	Pass	No action required
Numeric validation prevented partial upload	Pass	No action required
Valid attendance uploaded correctly	Pass	No action required
Correct success message displayed	Pass	No action required
Correct upload count displayed	Pass	No action required
Values stored as integers	Pass	No action required
Duplicate attendance prevented	Pass	No action required
Student could be deleted while attendance remained.	Fail	Add dependency validation to Student deletion so students with attendance cannot be deleted until attendance is
Attendance remained correctly associated with	Pass	No action required

Attendance remained correctly linked to student	Pass	No action required
Attendance remained associated with student ID	Pass	No action required
Current relationship was reflected correctly	Pass	No action required